

3 Algorithm

To find the distribution and other quantities of interest, we proceed in a manner similar to the constant rate case. In the algorithm below, we use the same notation for functions and their numerical discretizations in t . Functions of one time variable should be thought of as row vectors and functions of two time variables as matrices. We assume a partition of size n with $\Delta t = \frac{1}{n}$.

There are trade-offs in the algorithm between accuracy and speed of computation. In particular, When both the derivative and the adjugate are large, we compute $\text{Res}_{\lambda_k} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t_j - 1, t_j, \lambda) \right)^{-1} \right]$ using Equation (24) from Corollary 3. When the magnitude of these quantities is not a concern, we can compute $\text{Res}_{\lambda_k} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t_j - 1, t_j, \lambda) \right)^{-1} \right]$ using $a(\lambda_k) \text{adj} \left(\mathbf{I} - \widetilde{\mathbf{W}}(t_j - 1, t_j, \lambda_k) \right)$ as in Corollary 1 which will typically be faster.

It is faster to compute $\widetilde{\mathbf{W}}(t_j, t_{j+1}, \lambda)$ using the approximation

$$\widetilde{\mathbf{W}}(t_j, t_{j+1}, \lambda) \approx \exp \{ \Delta t (\mathbf{T}((j - .5)/n) - \lambda \mathbf{C}) \}$$

for $j = 1, \dots, n$, but more accurate to compute these quantities using an ODE solver.

1. Solve for $\gamma(t)$.
2. Find the zeros of $\det(\mathbf{I} - \widetilde{\mathbf{W}}(t - 1, t, \lambda))$. These will be the singularities of the LST. We choose one of two possible approaches. For either approach, we will find that there are infinitely many zeros. We call the zeros of this function, λ_k . Sort these zeros so that real zeros are first in descending order. Complex zeros will cluster near a few constant real values with uniformly spaced imaginary part. Sort these zeros in descending order of real part and within those groups by increasing value of imaginary part. The zeros will occur in complex conjugate pairs, so we only need to find those zeros with positive imaginary part. Based on Remark 3, we use a grid search method near the asymptotic estimates. We verify that we have all of the zeros by comparing the zeros we find with the asymptotic zeros.
 - (a) We can solve for the zeros of $\det(\mathbf{I} - \widetilde{\mathbf{W}}(t - 1, t, \lambda))$ (will have the same zeros for any t) directly or
 - (b) Define a function $\min[\text{svd}(\mathbf{I} - \widetilde{\mathbf{W}}(0, 1, \lambda))]$ as a function of λ and minimize it. Here svd is the singular value decomposition of the matrix $\mathbf{I} - \widetilde{\mathbf{W}}(0, 1, \lambda)$. For λ which are singularities, this value will be zero. Finding the zeros of this function is equivalent to solving for the zeros of $\det(\mathbf{I} - \widetilde{\mathbf{W}}(t - 1, t, \lambda))$ but is more numerically stable. See Remark 6. Recall that the determinant does not depend on t , so solving $\min[\text{svd}(\mathbf{I} - \widetilde{\mathbf{W}}(0, 1, \lambda))]$ is (up to numerical error) the same for $\min[\text{svd}(\mathbf{I} - \widetilde{\mathbf{W}}(t - 1, t, \lambda))]$ for any t .
3. Solve for $\mathbf{P}(t, 0)$. The row vector that represents $\mathbf{P}(t, 0)$ will solve the equations: $\mathbf{P}(t, 0) = \mathbf{P}(t, 0)\mathbf{H}$ and $\mathbf{P}(t, 0) = \varphi(t, 0)$. First we compute $\mathbf{H}$, and then we express $\varphi(t, 0)$ as a product involving $\mathbf{P}(t, 0)$.

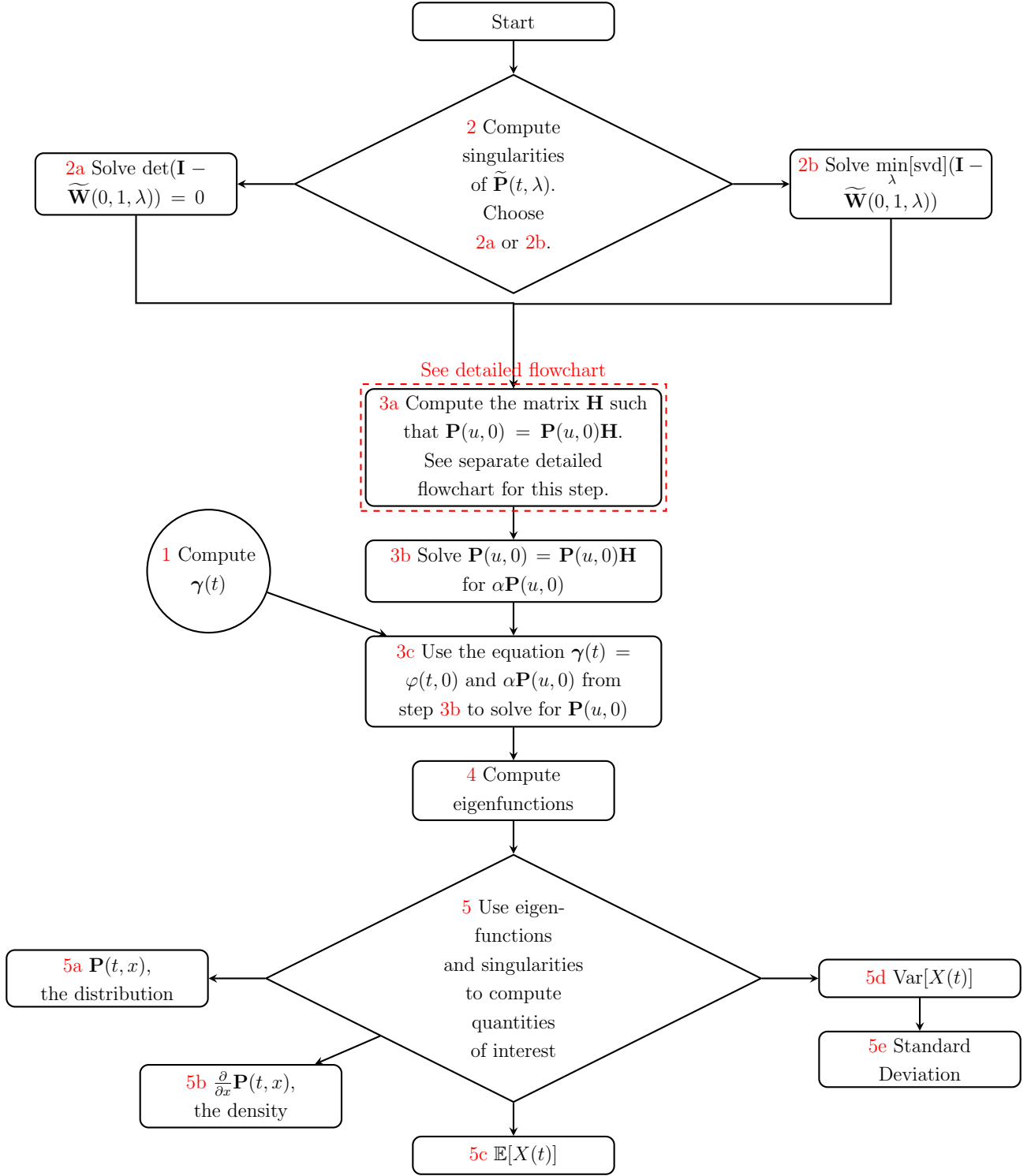


Figure 1: Algorithm flowchart

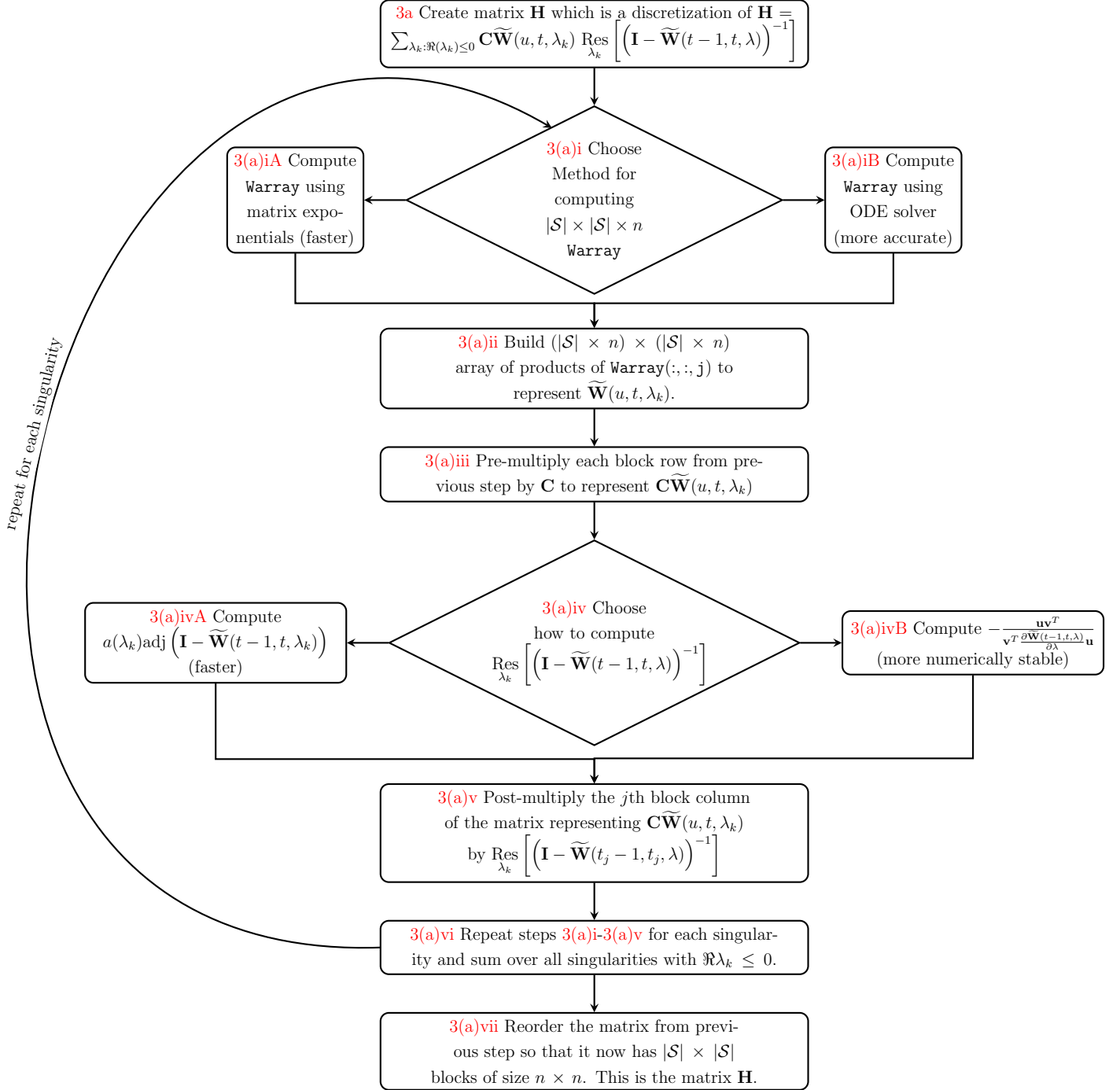


Figure 2: Algorithm detail for step 3a

- (a) Compute $\mathbf{H}$. First we will compute a matrix $\mathbf{A}$ that is organized in $n \times n$ blocks of size $|\mathcal{S}| \times |\mathcal{S}|$ blocks. $\mathbf{H}$ contains the same information as $\Delta t \mathbf{A}$ but is organized in $|\mathcal{S}| \times |\mathcal{S}|$ blocks of size $n \times n$. $\mathbf{A}$ is given by

$$[\mathbf{A}_{(i,j)}]_{(\ell,q)} = \left[\sum_{\lambda_k: \Re(\lambda_k) \leq 0} \mathbf{C} \widetilde{\mathbf{W}}(t_i, t_j, \lambda_k) \operatorname{Res}_{\lambda_k} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t_j - 1, t_j, \lambda) \right)^{-1} \right] \right]_{\ell,q}.$$

The formula inside the square brackets gives the matrix corresponding to initial time $t_i = (i-1)/n$ and ending time $t_j = (j-1)/n$ for $i, j = 1, \dots, n$. This step is based on Equation (43) in Corollary 4. This is an $|\mathcal{S}| \times |\mathcal{S}|$ matrix where $|\mathcal{S}|$ is the number of phases. The (ℓ, q) entry of the (i, j) block represents a transition from phase ℓ to phase q ($\ell, q \in \{1, \dots, |\mathcal{S}|\}$) when observed initially at time t_i and then again at t_j .

- i. Compute **Warray**. Compute a three dimensional array **Warray** with dimensions $|\mathcal{S}| \times |\mathcal{S}| \times n$. **Warray** $(:, :, j)$ represents $\widetilde{\mathbf{W}}(t_j, t_{j+1}, \lambda_k)$. We store $\widetilde{\mathbf{W}}(t_j, t_{j+1}, \lambda_k)$ in **Warray** $(:, :, j)$.
 - A. **Warray** will contain the matrix exponentials $e^{\Delta t (\mathbf{T}((j-0.5)/n) - \lambda \mathbf{C})}$ for $j = 1, \dots, n$ or
 - B. for greater precision, we can use an ODE solver to find $\widetilde{\mathbf{W}}(t_j, t_{j+1}, \lambda_k)$.
- ii. Build matrix of products of $W_j = \mathbf{Warray}(:, :, j)$. The product $\prod_{s=j}^m \mathbf{Warray}(:, :, s)$ approximates $\widetilde{\mathbf{W}}\left(\frac{(j-1)}{n}, \frac{m}{n}, \lambda_k\right)$. This is an application of the semi-group property of $\widetilde{\mathbf{W}}$. We will construct an $(n \times |\mathcal{S}|) \times (n \times |\mathcal{S}|)$ matrix composed of $n^2 |\mathcal{S}| \times |\mathcal{S}|$ blocks. Denote by $W_j \mathbf{Warray}(:, :, j)$, then we will build a matrix

$$\begin{bmatrix} \prod_{j=1}^n W_j & W_1 & W_1 W_2 & \cdots & \prod_{j=1}^{n-2} W_j & \prod_{j=1}^{n-1} W_j \\ \prod_{j=2}^n W_j & \left(\prod_{j=2}^n W_j\right) W_1 & W_2 & \cdots & \prod_{j=2}^{n-2} W_j & \prod_{j=2}^{n-1} W_j \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ \prod_{j=\ell}^n W_j & \left(\prod_{j=\ell}^n W_j\right) W_1 & \left(\prod_{j=\ell}^n W_j\right) W_1 W_2 & \cdots & \prod_{j=\ell}^{n-2} W_j & \prod_{j=\ell}^{n-1} W_j \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ W_{n-1} W_n & W_{n-1} W_n W_1 & W_{n-1} W_n W_1 W_2 & \cdots & W_{n-1} W_n \prod_{j=1}^{n-2} W_j & W_{n-1} \\ W_n & W_n W_1 & W_n W_1 W_2 & \cdots & W_n \prod_{j=1}^{n-2} W_j & W_n \prod_{j=1}^{n-1} W_j \end{bmatrix}.$$

Once we have the $(j-1)$ st block column, we obtain the next block column by post-multiplying it by W_j and replacing the j th block in the column with W_j . Block (ℓ, q) for $\ell < q$ of the block matrix is given by $\prod_{j=\ell}^{q-1} W_j$ and by $\prod_{j=\ell}^n W_j \prod_{j=1}^{q-1} W_j$ if $\ell \geq q$. This matrix approximates the function $\widetilde{\mathbf{W}}(t_i, t_j, \lambda)$ at each ordered pair of times within the period (t_i, t_j) for $i, j = 1, \dots, n$ (where for $j < i$, we use instead $(t_i - 1, t_j)$), and for each ordered pair of phases.

- iii. Pre-multiply the block rows of the matrix from step 3(a)ii by $\mathbf{C}$. The (i, j) block represents $\mathbf{C} \widetilde{\mathbf{W}}(t_i, t_j, \lambda)$.
- iv. Compute the residues $\operatorname{Res} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t_j - 1, t_j, \lambda) \right)^{-1} \right]$ and store them in an $|\mathcal{S}| \times |\mathcal{S}| \times n$ array, **Pdiag**. The j th entry, **Pdiag** $(:, :, j)$, stores $\operatorname{Res} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t_j - 1, t_j, \lambda) \right)^{-1} \right]$. We can compute these in one of two ways:

A. We apply Equation (24) from Corollary 3 to compute the residues. This method is preferred if both $\frac{d}{d\lambda} \det [\mathbf{I} - \widetilde{\mathbf{W}}(0, 1, \lambda)]$ and $\text{adj} [\mathbf{I} - \widetilde{\mathbf{W}}(0, 1, \lambda)]$ are large because it avoids catastrophic cancellation errors.

B. Alternatively use the fact that

$$\text{Res} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t_j - 1, t_j, \lambda) \right)^{-1} \right] = a(\lambda_k) \text{adj} [\mathbf{I} - \widetilde{\mathbf{W}}(0, 1, \lambda)]$$

as in Corollary 1.

v. Post-multiply the j th block column of matrix from Step 3(a)iii by the j th residue matrix, $\text{Pdiag}(:, :, j)$ for all $j = 1, \dots, n$. The (i, j) th block now contains

$$\mathbf{C}\widetilde{\mathbf{W}}(t_i, t_j, \lambda) \text{Res} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t_j - 1, t_j, \lambda) \right)^{-1} \right].$$

vi. Repeat steps 3(a)i-3(a)v for each singularity and sum over all singularities with $\Re \lambda_k \leq 0$. The (i, j) th block now represents

$$\sum_{k: \Re \{\lambda_k\} \leq 0} \mathbf{C}\widetilde{\mathbf{W}}(t_i, t_j, \lambda) \text{Res} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t_j - 1, t_j, \lambda) \right)^{-1} \right].$$

vii. Rearrange rows and columns, so that we now have a block matrix with $|\mathcal{S}| \times |\mathcal{S}|$ blocks of size $n \times n$ and multiply the resulting matrix by $\Delta t = \frac{1}{n}$ and return the $\mathbf{H}$ matrix.

(b) If we take $x = 0$ in Equation(21) in Corollary 1, we have $\mathbf{P}(t, 0) = \mathbf{P}(t, 0)\mathbf{H}$ and we can solve for a non-negative vector $\alpha \mathbf{P}(t, 0)$ where α is an unknown positive constant. Writing $\mathbf{H}$ in block form, this becomes

$$\begin{aligned} \begin{bmatrix} \mathbf{0} & \mathbf{p}_- & \mathbf{p}_0 \end{bmatrix} &= \begin{bmatrix} \mathbf{0} & \mathbf{p}_- & \mathbf{p}_0 \end{bmatrix} \begin{bmatrix} \mathbf{H}_{++} & \mathbf{H}_{+-} & \mathbf{H}_{+0} \\ \mathbf{H}_{-+} & \mathbf{H}_{--} & \mathbf{H}_{-0} \\ \mathbf{H}_{0+} & \mathbf{H}_{0-} & \mathbf{H}_{00} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{0} & \mathbf{p}_- & \mathbf{p}_0 \end{bmatrix} \begin{bmatrix} \mathbf{H}_{++} & \mathbf{H}_{+-} & \mathbf{H}_{+0} \\ \mathbf{H}_{-+} & \mathbf{H}_{--} & \mathbf{H}_{-0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{p}_- \mathbf{H}_{-+} & \mathbf{p}_- \mathbf{H}_{--} & \mathbf{p}_- \mathbf{H}_{-0} \end{bmatrix}. \end{aligned}$$

Note that for $\ell \in \mathcal{S}_0$, the $n \times n$ block $[\mathbf{H}_{(\ell, q)}] = \mathbf{0}$ since $c_\ell = 0$. Also, note that $\mathbf{P}_\ell(t, 0) = 0$ for $\ell \in \mathcal{S}_+$, so we consider just the phases $i \in \mathcal{S}_-$. We seek to solve

$$\mathbf{p}_-(t) = \mathbf{p}_-(t)\mathbf{H}_{--}.$$

Solve this for a non-zero $\alpha \mathbf{p}_-(t)$. Choose α such that $\int_0^1 \alpha \mathbf{p}_-(t) \mathbf{1} dt = 1$. In step 3c we will replace this estimate with the correct value for α .

(c) From Corollary 5 we know that

$$\mathbf{P}(t, \infty) = \boldsymbol{\gamma}(t) = \int_{t-1}^t \mathbf{P}(u, 0) \mathbf{C}\widetilde{\mathbf{W}}(u, t, 0) du \text{Res}_{\lambda=0} \left[(\mathbf{I} - \widetilde{\mathbf{W}}(t-1, t, \lambda))^{-1} \right].$$

So we compute the matrix $\mathbf{F} = \mathbf{C}\widetilde{\mathbf{W}}(u, t, 0) \text{Res}_{\lambda=0} \left[(\mathbf{I} - \widetilde{\mathbf{W}}(t-1, t, \lambda))^{-1} \right]$ and solve $\boldsymbol{\gamma}(t) = \mathbf{P}(t, 0)\mathbf{F}$. Because $c_i = 0$ for $i \in \mathcal{S}_0$ and $\mathbf{P}_i(u, 0) = 0$ for $i \in \mathcal{S}_+$, this expression depends on only the phases in $\mathcal{S}_-$. Substitute $\alpha \mathbf{p}_-(t)$ into this expression and solve for α .

4. Compute the eigenfunctions:

$$\begin{aligned}\varphi(t, \lambda_k) &= a(\lambda_k) \int_{t-1}^t \mathbf{P}(u, 0) \mathbf{C} \widetilde{\mathbf{W}}(u, t, \lambda_k) du \operatorname{adj} \left(\mathbf{I} - \widetilde{\mathbf{W}}(t-1, t, \lambda_k) \right) \\ &= \int_{t-1}^t \mathbf{P}(u, 0) \mathbf{C} \widetilde{\mathbf{W}}(u, t, \lambda_k) du \operatorname{Res}_{\lambda_k} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t-1, t, \lambda) \right)^{-1} \right].\end{aligned}$$

Follow the steps outlined in steps **3(a)i-3(a)vi** to compute

$$\mathbf{C} \widetilde{\mathbf{W}}(u, t, \lambda_k) du \operatorname{Res}_{\lambda_k} \left[\left(\mathbf{I} - \widetilde{\mathbf{W}}(t-1, t, \lambda) \right)^{-1} \right] \quad (51)$$

and multiply the resulting matrix by the row array $\mathbf{P}(u, 0)$ computed above. Save the result in a matrix. Each row will represent an eigenfunction with one eigenfunction for each singularity. We call this matrix of eigenfunctions $\mathbf{E}$. Note that the matrix computed in (51) was computed previously when we computed the matrix $\mathbf{H}$ so we could store the computations from the earlier step, but the amount of storage required is likely to be prohibitive because we would be storing multiple matrices of size $(|\mathcal{S}| \times n) \times (|\mathcal{S}| \times n)$.

5. Computing quantities of interest.

- (a) $\mathbf{P}(t, x)$. Let $\boldsymbol{\lambda}$ be a vector of singularities and $\mathbf{x}$ a vector of x values. Compute the outer product $\mathbf{x}\boldsymbol{\lambda}^T$ and define $\mathbf{B}$ via element-wise exponentiation:

$$\mathbf{B} = \exp^{\odot}(\mathbf{x}\boldsymbol{\lambda}^T) \quad (52)$$

with $B_{ij} = \exp(x_i \lambda_j)$. Multiply the resulting matrix times the matrix $\mathbf{E}$ of eigenfunctions. The resulting matrix $\mathbf{B}\mathbf{E}$, has rows corresponding to each x value, and column values corresponding to each time within the period.

- (b) $\frac{\partial}{\partial x} \mathbf{P}(t, x)$. The density is given by $\mathbf{B} \operatorname{diag}[\lambda] \mathbf{E}$. As above, rows correspond to each x value, and column values corresponding to each time within the period.
- (c) $\mathbb{E}[X(t)]$. The mean is computed by multiplying a column vector of $[\frac{1}{\lambda}]$ times the matrix $\mathbf{E}$, then summing over the phases. The result will be a row vector of means as a function of time within the period.
- (d) $\operatorname{Var}[X(t)]$. The variance is computed by first finding $\mathbb{E}[X^2(t)]$ and then subtracting $(\mathbb{E}[X(t)])^2$. To find $\mathbb{E}[X^2(t)]$ we find a column vector $-\frac{2}{\lambda^2}$ and multiply that vector times the matrix of eigenfunctions, $\mathbf{E}$. We sum this over all phases and then subtract the row vector of the mean squared where the square of the mean is computed term by term.
- (e) The standard deviation is the term by term square root of the variance.